



Beyond Self-Improvement: the Quantified Self

**A study on the institutionalisation of self-tracking in
a digital age**

by

Perle Fainas-Martin

under supervision of

Philippe Bonditti

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Abstract

This contribution presents a reflection on the institutionalisation of the self-tracking practices of the Quantified Self in society in relation to Foucault's principle of discipline, Deleuze's concept of control, and the ideas of neoliberalism. By analysing the normalisation of these practices in society, this essay explores how the methods of surveillance and normalisation specific to Foucault's disciplinary societies and Deleuze's control societies are re-articulated in the context of digital surveillance in a neoliberal era. This research highlights the tensions between the quest for individual autonomy promoted by the original Quantified Self movement and the imperatives of social conformity in a surveillance society with pre-established norms.

Keywords: Quantified Self, Self-tracking, Discipline, Neoliberalism, Surveillance

Résumé

Cette contribution présente une réflexion à propos de l'institutionnalisation des pratiques de *self-tracking* du *Quantified Self* dans la société par rapport au principe de discipline de Foucault, de contrôle de Deleuze, et les idées du néolibéralisme. En analysant la normalisation de ces pratiques dans la société, cet essai explore comment les méthodes de surveillance et de normalisation propres aux sociétés disciplinaires de Foucault ainsi qu'aux sociétés de contrôle de Deleuze sont réarticulées dans un contexte de surveillance digitale dans une époque néolibérale. Cette recherche met en lumière les tensions entre la quête d'autonomie individuelle promue par le mouvement original du *Quantified Self* et les impératifs de conformité sociale dans une société de surveillance avec des normes préétablies.

Mots-clés : *Quantified Self*, *Self-tracking*, Discipline, Néolibéralisme, Surveillance

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Introduction

In an article by the New York Times in May 2010, Gary Wolf, American writer and co-founder of the Quantified Self movement, said “Humans make errors. We make errors of fact and errors of judgment. We have blind spots in our field of vision and gaps in our stream of attention. Sometimes we can’t even answer the simplest questions ... We make decisions with partial information. We are forced to steer by guesswork. We go with our gut. That is, some of us do. Others use data.” (WOLF 2010) This data originates from individual’s self-tracking of their routine, such as one’s daily dose of coffee per day, or another’s number of hours slept per night. Although self-tracking practices have existed for centuries, for example with the writing of an individual’s journey through a personal diary, the growing feeling for self-tracking practices and individuals’ quantification in the 21st century led to a larger phenomenon: the Quantified Self. The appearance of the Quantified Self movement (in uppercases) took these quantified self practices (in lowercases) to a whole new level. Its creation by the two Wired magazine editors Gary Wolf and Kevin Kelly in 2007, followed by the creation of the Quantified Self website in 2008, reflects a growing fascination with the potential of technology to unlock a deeper understanding of ourselves. The slogan of the Quantified Self website indicates it well: “self-knowledge through numbers”. It sounds logical to say that you cannot improve what you cannot see, and therefore quantify, which explains the widespread adoption of objects connected not just to our homes, but also to our body, such as wearable fitness and sleep trackers. The popularisation of the movement led to a broader implementation of self-tracking practices into individuals’ daily life. Thanks to new technologies, people are effectively putting more care in their bodies by quantifying and evaluating themselves. The constant evaluation of oneself and its dangers seems one of the most recurring popular concerns as the Black Mirror episode “Nosedive”, a Netflix series, brought it out. After watching it, I started to wonder how we got there, and I decided to think about it and share my reflections to question the perceived increasing need for evaluation and quantification in modern societies.

Self-tracking is far from new. According to Deborah Lupton (2012), self-tracking “involves practices in which people knowingly and purposely collect information about themselves, which they then review and consider applying in their lives”. Self-tracking as a practice goes all the way back to the XVIth century with early self-tracking practices such as with the Italian physician Sanctorius of Padua, who is considered the father of modern quantitative experimentation in medicine. In order to study energy expenditure in living systems, he tracked his weight in comparison to his food intake and elimination for 30 years. Upon this comparison, he made major discoveries concerning perspiration and metabolism. From then, the practices of self-tracking persisted in the form of personal writings. For example, individuals could use personal diaries or autobiographies in which they applied to themselves instruments of control and management of the body, such as by making lists, balance sheets, or notes of diverse and various observations. However, it remained at an individual level, until the creation of the Quantified Self website, and then, the Quantified Self community. Consequently, the quantification was extended to aggregated data, where multiple quantified selves as self-trackers share and work collaboratively with their data.

The history of the Quantified Self phenomenon can be divided into four phases of the “laboratory of oneself”. During the first phase from around 2007 to 2011, Quantified Self was mostly

reduced to an “experimentation”. Not yet known to the wider public, the first adepts of the movement were mostly already passionate about technology and early self-tracking. A few platforms were created, such as MyFitnessPal in 2005 and Fitbit in 2007. For the first time, with the creation of the Quantified Self website and community in 2008, self-tracking has been extended to group data: multiple quantified selves share and work collaboratively with their data, in order to aggregate the numbers and obtain statistics to define the norms. The goal was mostly to optimise health wellness. The second phase, from around 2011 to 2014, marks the introduction to Quantified Self practices to the wider public and the movement became then known with the main goal of the optimisation of the self. The evolution of accessible and free applications popularised self-tracking practices, and its association to a larger culture of improvement of the self and productivity. Practices have been extended to multiple domains with the goal of optimising performances, like for example for work, sleep, or relations. Self-tracking practices then got more and more integrated into daily life as wearable sensors and mobile applications became omnipresent: this is the beginning of the third phase of quantification of life from around 2014 to 2016. Self-tracking practices have also begun to expand beyond the traditional realms of health and fitness, and individuals have begun using data to track their emotions, social interactions, and even spiritual practices. Finally, the last phase represents the institutionalisation of self-tracking practices, from around 2016 to today. With the normalisation of scores and responsibilities based on data, self-tracking is now widely used in diverse institutional contexts like in hospitals, schools and the workplace.

Quantified Self: a technological promise for human progress

There is no doubt in the literature that the evolution of the Quantified Self implies a growing feeling of a desire by individuals to improve their quality of life, and the advancement of technology being able to finally respond to these needs in a more efficient, direct and more detailed way. Numbers are accepted to be a way to move from uncertainty to certainty, which is defined by adherents of the Quantified Self movement as the main source of actual knowledge. (WOLF, 2010) Algorithms are accepted as a technology that is unable to lie, and even if numbers sometimes feel “cold and less resonant emotionally”, they are accepted to be more precise and detailed than anything else at helping individuals picking healthier habits and resolving life problems as they are considered rational, in contrary to the human mind. (Ibid.) This logic induces to an “hypertrophisation of logos”: from a form of pure rationality comes knowledge, and in this case, from the numbers resulting from the self-tracking of the individual comes the knowledge of the body by the individual. (CHIODO, 2022: 1667)

Moreover, the movement also marks a shift into the democratisation of the access to data: Big Health data streams are becoming increasingly consumer available due to multiple reasons, such as the evolution of computing technology and its diffusion and democratisation in many spaces of the social body, and accordingly the popularisation of self-tracking practices thanks to the development of new, more precise and more available technologies. There is a clear shift from data being strictly used for scientific practices, to a more universal and meaningful use by all individuals. It marks the beginning of a new and personalised health wellness stream. (SWAN, 2013) According to Melanie Swan, the Quantified Self represents “any individual engaged in the self-tracking of any kind of biological, physical, behavioural, or environmental information. There is a practice stance toward obtaining information and acting on it.” Individuals now have the capacity to go to doctors with many information about their body

already prepared. Numerous self-trackers have discovered workable answers in fields where the conventional healthcare system would never have looked for or used for their particular situation. Self-tracking serves as a mechanism of empowerment for the individual: these tools allow them to gain more knowledge over their body and mind, before being reserved to health professionals. With the exception that with the creation of the Quantified Self community, not only is the individual capable to track themselves in order to resolve or optimise a specific lifestyle issue ($n=1$), but multiple individuals are able to create an aggregation of each one's data in order to share and work all together to improve themselves. (Ibid.)

The concept of self continues to change, as individuals react to changes in culture and technology. Here, the quantified self activities represent a kind of new means of enabling constant and autonomous creation of the self. From then, the new wearable technologies of the Quantified Self could permit efficient and automated self-tracking in a way to transform the self by adding an extended connected self. These technologies can be understood as exoself components, a vision of the extended human in the future. (BOSTROM & SANDBERG, 2009) The exoself can be understood as a combination of new senses defined as exosenses, in order to facilitate everyday habits. Wearable components such as watches and phones can be understood as so, as well as more precise exosenses such as augmented reality glasses, haptics (touch-based) or the Memex project (memory augmentation of experiences by artificial intelligence). (SWAN, 2013) As new technologies continue to emerge, humans are becoming more and more dependent upon it for a lot of different aspects, including interactions with the outside world. However, in the meantime, technology is increasingly able to provide a richer, more detailed, and more controllable relationship with the world with the exosenses and the Quantified Self in general. The Quantified Self phenomenon is believed by the members of the Quantified Self community to be a mean to go towards a more advanced future society, as it increases the following human commodities: choice, understanding, consciousness and freedom. It does not take any decision at our place, but rather, increases our comprehension of the world to be more optimised than our said simple human subjectivity. (Ibid.)

Datafication / Dataism

The Quantified Self movement emerged in a context of datafication of society. Datafication can be understood as the transformation of all aspects of human life into data, from work performance to emotions and personalities. It is in the same way that personality tests used by HR for the recruitment of employees are being replaced with data obtained from mobile phones or social media, or that identification devices are being replaced with passwords. Once something is datafied, it is possible to transform its original purpose and find new values to these informations, such as in this example with the potential recruitment of an employee through his social media. (CUKIER & MAYER-SHÖNBERGER, 2013) Here, the Quantified Self is seen as a mere characteristic of the larger process of datafication of the world. (RUCKENSTEIN & SCHÜLL, 2017)

Moreover, there exists an ideological aspect to datafication: *dataism*. It refers to the growing belief in data as a representation of human experience. The popularisation of the datafication movement throughout society finds its roots in this belief. Dataists consider that data can actually represent social life better and more objectively than humans, also called pre-digital interpretations. According to Yuval Noah Harari in his book *Homo Deus: A Brief History of Tomorrow*, dataism can be interpreted as an emerging ideology or even a new form of religion,

on which the entirety of human species could be interpreted as a “single data processing system, with individual humans serving as chips”. (HARARI, 2017: 428) In order to maximise dataflow and for the algorithms to be more precise and rational, dataists need to connect even more data together. He concludes that the only logical conclusion of this never-ending flow of growing data is that eventually, humans will give algorithms every authority in their lives. For dataists, the wellbeing of humanity depends on the freedom of information. One of its first victims is considered to be Aaron Swartz, a 26-year-old man who committed suicide in 2013 to escape facing prison. He was on trial for hacking computers in order to try to make academic papers available to everyone and make the freedom of knowledge accessible. Here is the main reason as to why most scientific articles in JSTOR today are free to access. Thus, making a free data flow the ultimate key for the wellbeing of humans for dataists. (Ibid.)

The incentive to discipline

Michel Foucault is a frequent reference in the self-tracking literature review. Most concerns about the quantified self are related, in some forms or shape, to Foucault’s research themes. From one part, they seem to embody a specific form of the “techniques of the self” and “bio-power”, as he noted that discipline of the body and the regulation of the population are the two ways in which the organisation of power on life got deployed. (FOUCAULT, 1975) From another part, they also operate in similar ways to the Panopticon. Designed by Bentham, this prison plan works as a model for the way society should function in order to maintain discipline, by the individuals needing to believe that any person could be watched at any time. This way, individuals would just police themselves. (Ibid: 202-203) And this model can work for many different situations, in hospitals, education, workplace, prisons, etc. Whenever there are multiple individuals upon which a certain behaviour must be imposed in order to keep society disciplined, the panoptic schema can be used: the proliferation of this model is what Foucault conceptualise as the disciplinary societies. (Ibid: 205)

Surveillance processes can be seen as more complex than the Panopticon allows. Three types of surveillance relations can be analysed: face-to-face, bureaucratic file-based, and electronic interface — they can also be found together. In the case of the Quantified Self, it mostly relates to the third one. (LYON, 2006) A key theme to surveillance in modern society is the watched subject’s active participation in the process of surveillance. First, the subject does not just accept its assigned role like it would in a prison, as humans are not exactly like robots. They are subjected and disciplined individuals to a certain limit, as each individual is different from each other and constantly reconfiguring its identity. Furthermore, subjects often actually participate in their own surveillance and are the same ones trying to implement these surveillance systems. (Ibid)

Since the end of the Second World war, there has been a slow shift from the disciplinary societies to “control” societies, caused by the technological advancement and the infiltration of capitalism into every aspect of life. As time passes, closed systems operating over time present in disciplinary societies shift to much more fluid, rapid and constant ways of doing things. For example, the Panopticon that was assuring a strict surveillance, for a long time and in a closed space, gets slowly replaced by a constant control in everyday life. Before, prisoners used to only have one choice: go to prison for a certain time, with guards watching over them for a certain time. Now, there have been more and more what we call “alternative” sentences,

like an electronic bracelet, or sentences in the individual's home through a computer. (DELEUZE, 1990)

From discipline to control, there is a shift from behavioural to categorial issues. As the data produced by the Quantified Self applications can be used as some forms of control based on social sorting, they can then be conceived and used as surveillance apparatus. Digital surveillance in a universe of data can facilitate evaluation and categorisation of individuals, and so be categorised by a "categorial suspicion". In the same way, insurance companies could be working with self-tracking applications oriented toward health wellness, by for example increase the pay of those doing a certain numbers of steps per day. (MCFALL, 2020) Indeed, not only control societies have an impact on the spatial dimension, but also on temporality. In disciplinary societies, everything was constantly re-starting: from school to house, from house to factory, and inside each of the institutions. But since evaluation processes are now everywhere and constant, everything is in continuum. If disciplinary process were about norms and conducts, control processes are now about life itself. (DELEUZE, 1990)

The making of the Quantified Self subject

The same way control processes are about life in itself rather than a set of norms and conducts, the neoliberalism movement embodies these principles by considering the individual as an end to itself. Neoliberal governmentality produces individuals and makes them consider themselves as "entrepreneurs of the self". It pushes them to make rational calculations of the costs and benefits of each choice they make, in order to maximise the returns on their efforts. (FOUCAULT, 2004: 232) However, while capitalism generates a constant revolution of the economic order, the individual does not just spontaneously adapt to this logic. This way, there is a need for politics centred around the individual and its social life. So, there is a need for a new production of subjectivity: a subject based on the internalisation of the logic of the market or "entrepreneurial governance". (DARDOT & LAVAL, 2014: 187)

In order to make the individual internalise this logic, there is a need to give value to work. This value, the evaluation, is the new principle to administrate and measure the value of individuals. (BOUSSARD, 2009) The contract that an employee signs, mechanism of the neoliberal logic of individual evaluation, does not only concern the employee with its superior anymore: it concerns the employee with society and creates a new type of relation in society; the agency relation. This relationship represents the heart of the company, and from then results the organisation of work by agents of the evolution of individuals. (JENSEN & MECKLING, 1976) The evaluation also serves as a mean of legitimation of the actual structures and the presence of the management by numbers, pretexting a "logos gestionnaire" or logic and rationality of the management involving the diversity of Foucault's "jeux de pouvoirs" involved in the construction of the managed subject. In order to achieve and maintain this level of management and evaluation over the individuals, the need for the "dispositifs" of power theorised by Foucault are essential. (BOUSSARD, 2009: ch8)

Contribution

Nowadays, it is possible to get to know yourself differently and potentially improving what you found out thanks to new wearable and trackable technologies of the Quantified Self. However, understanding this phenomenon requires a nuanced approach, that also considers the role of two key concepts: disciplinary power, and the impact of neoliberalism. This growing fascination to quantify ourselves exists within a broader societal trend of datafication and evaluation. From birth, to education, to healthcare to the workplace, individuals are subject to metrics and assessments. The modern individual is born with a pressure to optimise performance in various domains, and the line between personal wellness and external pressures is becoming increasingly blurry. Existing scholarship has explored the Quantified Self through these lenses, often in isolation. Some explains how Michel Foucault's concept of disciplinary societies highlights the way self-tracking technologies internalise control mechanisms and thus shape individuals to conform to existing norms. Others emphasise how the movement aligns first and foremost with neoliberal principles of self-awareness and self-optimisation, in order to conform to the idea of the entrepreneur of the self.

Therefore, my research question is: in a society increasingly digitalised and data-driven, to in what ways does the Quantified Self movement, originally intended for self-knowledge and improvement, actually contribute to a neoliberal culture of self-surveillance and internalised control?

This thesis argues that the understanding of the role of the individual in the Quantified Self can emerge by the understanding of the connexion between these two concepts. Foucault's concept of disciplinary power reminds us that power mechanisms are not neutral: considering quantified self practices in the light of today's society of evaluation as power mechanisms, they are not neutral from the way they are designed, the data they take, to the algorithms that gives potential solutions to improve a specific issue in the individual's lifestyle in order to push productivity. In this sense, the Quantified Self can be seen as a contemporary manifestation of disciplinary power, that operates mostly through internalised norms rather than external constraint. In the meantime, neoliberalism, with its emphasis on self-improvement and the encouragement for individuals to consider themselves as entrepreneurs, creates a fertile ground for the popularisation of the movement of the Quantified Self.

To develop this research question, I will adopt an analytic lecture of a corpus of scientific texts from disciplinary and control mechanisms, broad surveillance, neoliberalism and existing Quantified Self literature. Through content analysis, I aim to observe and highlight the assumptions with which power mechanism of the disciplinary societies within which Quantified Self practice thrives, as well as the link between neoliberal principles and the motivations pushing an individual to use self-tracking practices. I will base most of my corpus on the classes at the Collège de France and the book *Discipline and Punish* (1975) of French philosopher Michel Foucault, regarding power, surveillance and neoliberalism. Regarding the limits of my contribution, I decided not to focus on the subject of anonymity in digital experiences and the ulterior uses of the self-tracking data by companies and other actors, as I decided to take a more philosophical approach to the subject. In the same way, I chose not to develop the discourse regarding the different definitions of neoliberalism, to have more scope of expression on the subject of discipline. Throughout this thesis, I will consider the Quantified Self in upper-cases, talking about the movement created by Gary Wolf and Kevin Kelly and its influence in today's popularisation of self-tracking wearables and applications. Therefore, it corresponds mostly to the last phase of the laboratory of the self from around 2016 to today (2024), with the institutionalisation of these practices into everyday life through applications but also into

diverse institutional contexts like schools, hospitals, and the workplace. In addition, this essay considers a large panel of individuals who use self-tracking practices: from the average person just trying to lose weight through a free downloaded application on their phone, to the “hard quantifier” member of the community using daily many wearable tools and monitors.

In the subsequent questions, I will begin by looking to understand how the individual adopts a stance of governance towards themselves in digitalised societies with the self-tracking practices, from disciplinary societies to the societies of control. Then, I will try to understand the neoliberal implications of the making of the subject in the Quantified Self era, such as with the rise of the figure of the entrepreneur of the self and the push for the ultra-individualization in order to keep the individual from acting upon anything else that could not be commodified.

Part I. Self-monitoring & self-surveillance in digitalised societies

Each aspect of the lifestyle promoted by Quantified Self movement of Gary Wolf and Kevin Kelly has provoked opposite reactions, sparking a debate revolving around some basic conceptual dichotomies: empowerment versus surveillance, self-awareness versus reductionism of the human life. First, this thesis will detail the existing theories about discipline and surveillance through Foucault’s disciplinary societies and Deleuze’s control societies, and their correlation with the Quantified Self.

1. 1. Foucault’s disciplinary societies

In his book *Discipline & Punish* (1975), Foucault details how public torture disappeared and got replaced by the apparatus of punishment and surveillance behind the walls and the bars of the prison. In order to grasp Foucault’s concept of discipline, it is necessary to comprehend the genealogy behind its conception. In the first part of the book, “Torture”, Foucault displays the shift between how public torture was exerted before as a punitive scandal in order to show the power of the royalty and to impress the crowds, and how progressively through the 17th century it began to be viewed unfavourably and considered as a mere useless repetition of the horrors of the committed crimes. From then emerged a need for a change of the relation between the body and the punishment. Instead of torture, the individual is now submitted to work. It serves as a double purpose, as on top of receiving the punishment and being corrected, the individual is also useful to society. Foucault calls this new era the age of sobriety in punishment. By making the individual useful to society and more inclined to discipline by the punishment, a new political economy of the body is created: this double use of the individual should be considered the basis of the political order to reach social efficiency as a society. So, more than the creation of prisons to lock individuals away, it is on this model than an entire disciplinary society is created to organise and frame the lives of individuals. In this society, the power is not centralised and localisable but operates under the form of a network that multiplies under the form of ‘micropowers’. These diffuse micropowers constitute what Foucault calls the ‘microphysics’ of the body, that allow the subjugation on the closest parts of an individual, from its life, habits, relations, to his own body. From its birth in a hospital, its education in schools and universities, its work, prisons, asylums... All these places are called by Foucault “disciplinary institutions”. In order to keep the population from rebelling and for these institu-

tions to create “docile bodies”, they must be able to constantly observe and record the disciplined bodies and ensure the internalisation of the disciplinary mechanisms within the bodies being controlled. To achieve these controls, they operate through various techniques such as: normalisation, hierarchisation, surveillance and sanction. What Foucault calls discipline can then be understood as the technology of power by which individuals are self-disciplined by incorporation of the disciplinary mechanisms. So, even if the prisons are the most visible aspect of disciplinary societies, it is actually the entire society that is concerned by these discipline mechanisms: from school to hospitals or factories and many others. The prison is merely a model to the one of the disciplinary societies, that aims to surveillance and subjugation of the individuals.

The most representative model of the disciplinary society is the prison model. In these places, a hierarchy surveillance is organised in which every individual is monitored at all times, including the prison supervisor supposed to monitor the prisoners. The objective is to correct, discipline and normalise these behaviours. In disciplinary societies, there is a power imbalance: those executing the surveillance, and those under surveillance. However, those executing the surveillance do not use demonstrations of power to force the subjects into behaving. It is the presence of observation that scares subjects into submission. Rather than the admitted presence of observation, it is the fear of a potential surveillance that forces individuals to submission. The architectural system of the Panopticon is a great example of this precise fear. Created in the 18th century by English philosopher and social theorist Jeremy Bentham, the Panopticon is a representation of an institutional building with an inbuilt system of control. Conceived as a prison, the plan is equally applicable to hospitals, schools, sanatoriums, and asylums. The architecture is the following: it consists of a rotunda, with an inspection house at its centre. Those executing the surveillance in the inspection house are able to watch the inmates. However, those under surveillance are not able to see when they are being watched: potentially, they are, but they have no mean of knowing when. Consequently, prisoners always are under the impression that they are being monitored, even if they are not. Therefore, they finish by monitoring themselves, and to integrate themselves and in themselves the surveillance and discipline. That is how the power can work in an automatic way, as it got integrated into bodies in itself, that became docile by the potential surveillance. According to Foucault, the Panopticon is the most accurate form of reflection of what disciplinary societies are, and the mechanisms of micropowers in it that insure efficiency and profitability. But every disciplinary institution resembles the prison model in a disciplinary society: schools, hospitals, factories, asylums, etc. Foucault even describes the disciplinary society in his last chapter of the book as a prison-city, where each individual is a disciplinary individual, that exert surveillance on themselves as well as the other.

The model of the prison, coercive and solitary, became the privileged model for sanction instead of the popular public torture scenery, representative and collective, for the reason that was stated before: the goal is to make the bodies obedient and compliant, and furthermore, to produce a coercion on the individuals that is never-ending on time and space. Discipline also requires an art of repartition of tasks: actives must be ensured by the timetable, every movement must be timed, body and health examinations must be multiplied, everywhere and every time, and a trace of the examinations must be kept. Discipline is not only about dividing and allocate the bodies, but to create micropowers to obtain subjugation of the individuals, under the impression that they get something out of it: “knowledge of their body”. It is precisely the knowledge on the body that allows the relations of power on the individual: for workers in

factories, children in schools, people considered as mad in asylums, etc. It is by this relation between knowledge and power (“savoir-pouvoir”) that bodies can become profitable for the economy: by knowing the body, it becomes useful by its economic efficiency in society. As evoked before, disciplinary societies need a relation of power and knowledge, so that individuals behave according to society’s truth and become disciplined to not be excluded from society. This relation between power and knowledge is normalised and enforced by what Foucault calls a “regime of truth”. The regime of truth comprises the dominant discourses, ideologies, beliefs, values, and norms that are promoted, enforced, and upheld by disciplinary institutions and authorities. Individuals are expected to internalise, adopt, and adhere to the societal truth, conforming to the prescribed norms, standards, and expectations. The goal is to make everyone adhere to a common truth, in order to discipline the population into following precise norms. In his lecture at the Collège de France named “Security, territory and population” on the 25th of January 1978, Foucault details what he calls norms, and the difference between normation as the establishment of norms and standards, and normalisation as the process of enforcing conformity to those norms within society. Therefore, the operation of disciplinary normalisation consists of trying to make people conform to the normal model conformed to the norms, and the normation refers to the used disciplinary techniques.

Here lays the Quantified Self, said to be a manifestation of the disciplinary power exerted in society. To confirm this theory, it is necessary to compare in the following section the disciplinary mechanisms to the Quantified Self mechanisms.

1. 2. The Quantified Self: a mechanism to enforce discipline

Beyond the incentive for self-knowledge, the Quantified Self uses mechanisms of surveillance to achieve the goal of the individual. According to Gary Wolf, this surveillance only serves greater purpose for the enhancement of the individual — Nevertheless, it is still a constant surveillance of the individual, through applications and wearable devices, and some can say that the Quantified Self repeats the mechanisms of the Panopticon, through different points. First, there is the potential risk of being watched by the guardian in the prison in the Panopticon. In the Quantified Self, this potential risk of being watched is not caused by the presence of a guardian of a prison, but rather by society in itself pushing norms into individuals through normalisation. The risk present here is the one of being excluded from society: the risk of being considered an “abnormal”. In order to avoid this possibility, which basically means “death” in society, the individual is pushed to conformity. This potential risk of being watched then turns into an impression of constant surveillance and judgment by the other individuals, even if they are not. Am I sportive enough, compared to the others? Am I mentally healthy enough, compared to the others? There is a fear and despise of people for being not healthy enough compared to society’s norms: from being too fat or too skinny compared to the norms, having an unstable mental health or being mentally sick in any way, or for working too much or not enough. Work is also a major part of health wellness, though this connection to work will be approached in the second part of this thesis. But this fear and apprehension about not meeting societal norms regarding health wellness is deeply ingrained into the main concepts of the Quantified Self, individuals constantly finding themselves questioning whether they fit through the norms or not. Journalist David Pogue even considers the Quantified Self as a system of “fitness through shame”, with the possibility of adding relatives into our health wellness applications and systems. (POGUE, 2013) Aesthetic norms are especially striking in this case, as it is easier for its peers to understand if an individual fits through them or not, and to extract

knowledge from it : “Those intentional and voluntary actions by which men not only set themselves rules of conduct, but also seek to transform themselves, to change themselves in their singular being, and to make their life into an oeuvre that carries certain aesthetic values and meets certain stylistic criteria.” (FOUCAULT, 1990: 10-11) Or, in another case, it is even possible to extract knowledge out of an unborn human, as it is now possible to know during pregnancy if a baby is suffering from Down syndrome, and to potentially abort in order to make the lives of the parents easier — or to avoid judgment from the others. Not respecting the health norms can also results in concrete forms of exclusion, further than a social exclusion: for example, prior to 2018, people diagnostic with the autism spectrum disorder could not apply for permanent residency in Canada. Banks can also legitimately refuse to give out credits to people that have a medical history of mental illnesses, diabetes, cancer, or other health reasons. Consequently, by integrating these norms and using the Quantified Self to have a personalised analysis on how to “improve” themselves and therefore meeting society’s standards and recede from being excluded, individuals finish by monitoring themselves, and integrating themselves and in themselves the surveillance and discipline, in the same ways as the mechanisms of the Panopticon operates. By making a representation of themselves accessible for individuals through the data of the Quantified Self tools outside of their own lived experience, it makes individuals vulnerable to behavioural norms and make them integrate them through the evolution of their Quantified Self practices. According to Dagiral, the experience of having this outside representation of ourselves makes the individual feel “guided” by the self-tracking data, or moreover, disciplined. (DAGIRAL, 2019)

This exertion of the micropowers implementing norms consists of the same mechanisms: through the body of the individual. In order to make the bodies compliant and docile, discipline must be exerted at all times through time and space. Through the Quantified Self, or more largely through new technology, surveillance is exerted at all times and everywhere — even for the non-compliant. Whistleblower Edward Snowden claims that everyone is under surveillance now, and that individuals now walk with a surveillance camera (referring to mobile phones) in their own hands at all times. (SNOWDEN, 2019) The Quantified Self follows this same principle — or even further, as the individual is fully aware of being monitored as it seems like it is a “choice” of surveillance to them. According to Laurence Barry, the real novelty of the Quantified Self compared to other disciplinary mechanisms is that the individual is being shaped, governed, and constituted by *algorithms* that analyse the individual, interpret the data and generate insights from it. Normally, the individuals would produce the truth, that is then interpreted by the “masters of truth”: for example, in schools, the student will produce “truths” through writings and exercices, while it is the teacher’s job to interpret these truths to determine their rightfulness compared to society’s norms. But here, the discipline is replaced by a code of conduct, freely chosen in appearance, by a subject to make sense of his own self. Thus, the individual is then produced by a discourse of truth that is the output of an algorithm. Laurence Barry establishes the distinction between the modern subject, developed with the injection to conform a certain narrative that has to be *said*, from the Quantified Self subject, that is driven by a “series of fluctuating numerical indicators that are immediately collected by sensors”. Indeed, even certain elements that used to be understood through said words like mental healthiness or feelings are now inferred from wearable technologies or online posts. (BARRY, 2020) Furthermore, dataist or hard followers of the movement even seem to consider the representation to the self without algorithms meaningless, as they fail to give you an “objective” perspective.

But are data and algorithms objective? The answer is obviously no, or at least, not completely. First, data is not neutral: it is shaped by the processes of collection, interpretation, and analysis, and through each stage can be introduced biases, as individuals are at the basis of their creation. Moreover, the metrics are yet still widely unable to quantify subjective experiences such as human emotions and feelings. Most applications related to mental health like MoodKit, Happify, MindDoc, and many others require the individual to write through words in a journal in certain designated parts of the application, as feelings can hardly be reduced through numbers (as of today). The same goes for algorithms: while they are designed to process data and make potential predictions about it, they may produce flawed or biased outcomes. The algorithms reflect the assumptions, values and priorities of their creators, which may not align with the experiences of the individual being quantified : the choice of which part of the data to take, how to interpret it based on specific current medical studies and which studies to take inspiration from (related to specifics in health wellness, many studies contradict others), what result to give out to users, or under what form, and many other factors. However, that does not make Laurence Barry's idea of the subjectivity of the self-based on the output of the algorithm as a novelty from disciplinary societies false : it is just the experience that is changed, to the integration of the norms through common understanding, education and *said* speeches, to the integration of the norms through a more digital experience, that may seem more cold and emotionless but appears to be more rational to users.

1. 3. The quantified subject: a representation of Deleuze's dividual

However, in his literature review about the Quantified Self, Éric Dagiral agrees that the model of auto-surveillance exerted by the Quantified Self finds its place in the continuity of surveillance like the one described in the Panopticon, but also admits that this continuity presents limits. Indeed, in Foucault's description of the disciplinary societies, the discipline exerted over the bodies is usually referred to disciplinary institutions like schools, hospitals, factories, and prisons that are characteristic of the disciplinary societies. However, the Quantified Self presents a difference: the self-tracking mechanisms, said to be understood as disciplinary mechanisms, do not have limits in time and space. On this basis, these mechanisms seem to participate as well to the description of the control societies described by Deleuze. (DAGIRAL, 2019) In the control societies, the mechanisms are being exerted in a way that is continuously and flexible, through institutions that are "gaseous" in the sense that power and responsibility are constant, and mostly, diffuse. In disciplinary societies, there exists a broad social clivage between the individual against the mass. However, in the control societies, the individual is fragmented within itself, to make what Deleuze calls the "dividual". The dividual refers to a version of the individual that is fragmented and represents the aggregation of divided proprieties — in the case of the Quantified Self, the dividual is the *me* resulting from the collection of all the data sets to which I belong. There is an incentive to transform individuals into a multiple set of numbers, and to make them become *quantifiable* entities. According to James Brusseau, the change becomes visible by the shift between the frequent use in the past of our identification that allows us to drive cars, get into bars, confirm the credit card as ours at the end of the night, to the more recurring use today of passwords, to trigger the debit card, read our mails, or connect to our different accounts : the individual becomes numbers, a fragmented dividual. (BRUSSEAU, 2020) This shift in the nature of discipline and surveillance to control reflects broader societal changes, such as the growing concept of self-management in neoliberal ideology.

Moreover, in control societies, there is a shift from behavioural to categorial issues. The data produced by the self-tracking tools can be used as a sort of digital surveillance, and therefore induce forms of control based on categorial social sorting. This data allows the classification of individuals into various digital groups, hidden, based on criteria such as health status, life-style choices, and consumer preferences. Then, this categorisation can be used to exert forms of control and influence over individuals by shaping their digital access to different resources, like with insurance companies to measure an individual's health risk or for employers to analyse an employee's productivity metrics. Moreover, after the 2022 decision of the Supreme Court in the United States, plenty of people called for a massive un-downloading of period-trackers applications, in order to avoid data being collected from these applications and thus a potential penalisation of anyone considering an abortion. As this type of application can meticulously show when period stops and starts and when a pregnancy stops and starts, the collected data could represent a real danger. This categorial sorting guides the individual into a more personalised digital experience, as algorithms will often show recommended ads from people with a similar profile. Thus, the creation of the digitalised individual, notably through the Quantified Self, makes the individual more fragile and easily influenced as each one of its interests is decrypted and utilised to maximise profit, and makes it easier for control mechanisms to exert power.

The sole analysis of the Quantified Self through the lenses of Foucault's disciplinary societies is insufficient to understand the scope of the phenomenon. From there, Deleuze introduces the control societies, for which neoliberalism takes a large part for the cause of this shift. Neoliberalism introduces a broader socio-economic context that shapes and influences the ways individuals engage with self-tracking and self-monitoring technologies.

Part II. Neoliberalism & the commodification of human experience

2. 1. The market-driven logic of Neoliberalism

Neoliberalism is a type of policy model, that incorporates both politics and economics the same way it can also be understood as an ideology, which places at its heart the institution of the market. It emerged in the mid-20th century, following the many interventionist policies of the welfare state in the United States. In the 70s, after many years of nationalisation of companies, several waves of privatisation of business appeared ensuing the growing feeling of a need for more liberalism: this is where neoliberalism, as an ideology, was said to be born. Neoliberalism includes several key principles, such as: free trade, as goods and services should be able to move as freely as possible between countries without taxes to promote competition, limited government intervention, with the government playing a very limited role in the economy to let the market freely move by itself, and individual liberty, as individuals should be free to make their own economic choices without government interference. At its core, the market seeks profits driving economic growth, productivity, and innovation. Margaret Thatcher, Prime Minister of the United Kingdom from 1979 to 1990, provided the clearest formulation of this rationality: "Economics are the method. The object is to change the soul." By promoting and encouraging liberty, neoliberalism promotes the concept of Homo Economicus, or the "economic man", as an idealised individual whose logic follows the one of the market: rational,

utility-maximising, who follows self-interest, and mostly is entrepreneurial. It is about encouraging individuals to have an entrepreneurial mindset, and to lead the individual into maximising their potential, productivity, and performance. This idea of the rational individual that considers themselves as entrepreneur of their own body is called by Foucault the “entrepreneur of the self”. As they invest in the pursuit of these skills, they usually develop more flexibility, adaptability, resilience, and versatility, which advantages the market flows. In order to push the flexibility of the market into individuals, the individual becomes a finality in neoliberalism, whereas it is rather considering a mechanism in disciplinary societies. The notion of “human capital” is defined in the Oxford English Dictionary as “the skills the labour force possesses and is regarded as a resource or asset”. This notion corresponds well to the ways self-trackers apprehend their experiences: to put into light the causes that affect them, in order to finally get the possibility to take control over them and maximise their personal capacities. The logic of the market behind this promotion of a better human capital suggests that individuals who possess great skills are more likely to have better job opportunities, thus earning higher incomes and achieving a greater career advancement in the competitive market. As a result, the individuals are seeking to pursue education, do more training, in order to upgrade their human capital. Moreover, when individuals upgrade their human capital, they usually aim to experience an increased income. This means that they gain more purchasing power, and as a result more consumption opportunities in order to enhance their quality of life and to take care of themselves. What they gain by upgrading their human capital and working is being reinvested into the economy of the market. This concept is directly linked to Deleuze’s concept of control, defined as a social organisation without spatial divisions and explicit prohibitions, that sparks the following question: where can the control come from if nothing is forbidden like in disciplinary societies? And the answer lies in data-driven marketing and social media strategies that regulate through incentives. Consumers seem to not be forced to do anything, but they are lured by the marketing appeal to exploit personal information — what Deleuze calls the “joys of marketing” (DELEUZE, 1990) that drive the control economy.

The role of the Quantified Self in this logic is the following: by allowing individuals to gain insights on their capacities, strengths and weaknesses, they are able to develop an image of self-awareness on themselves. This development allows them to regulate their skills through an active engagement with their own self-tracked data. Individuals are then able to enhance their productivity and flexibility, with the same goal of upgrading their human capital to achieve what other individuals and the norms expect them to be, and hence to become an entrepreneur of the self and have full control on the management of their body. Furthermore, the Quantified Self movement promotes a culture of lifestyle optimisation: the more people are transforming their everyday moves and thoughts into data, the more the algorithms can give them precise and helpful answers to solve specific lifestyle problems, and therefore the more individuals can improve themselves to be more efficient in the workplace and contribute to the flows of the market. All of the norms that the individual should reach for that become normalised through the disciplinary and control societies such as health wellness or beauty norms imposed by society, play a huge role into the induction of self-monitoring practices of themselves. The mechanisms of the Quantified Self become a mean for the individual to consider themselves as an entrepreneur of the self: or, furthermore, to transform the *quantified* individual into a *qualified* individual, using the same terms as Melanie Swan. (SWAN, 2013) The qualification is to be understood as the enhancement of an individual’s skills in order to correspond to what society is expecting from him.

2. 2. Society of evaluation

The incentive to quantification tools to measure ourselves is enforced by the societal shift towards what can be termed as the “society of evaluation”, in which individuals are constantly subjected to evaluation and comparison across various domains over their life, including education, work, health, or even social interactions. This system is implemented in an individual’s life as early as in education, to prepare the student’s employability, to the never-ending and continuous cycle of work in today’s control societies described by Deleuze. The use is to evaluate, interpret, and classify individuals through numbers. According to Angélique Del Rey, these methods of evaluation became omnipresent in society after the introduction of the New Public Management (NPM) method in the Anglo-Saxon public policy evaluation model. And since then, these methods of constant evaluation are not being questioned and are today considered the norm. Effectively, she calls this phenomenon the “tyranny of evaluation”: for her, the implementation of this system as the norm is founded on its claimed objectivity, and it is exactly what makes it dangerous. The evaluation numbers exert power on the entirety of an individual’s life and social life, but disguise themselves as “informations”, or even “truth discourse”. (DEL REY, 2013) Obviously, this formulation refers to Foucault’s principle of the regime of truth, that was evoked in prior parts. The disciplinary societies aim at forcing an identity and true norms into an individual and correcting his behaviour — this evaluation frenzy is one of its mechanism, through considering individuals by numbers as it is the case with the Quantified Self may be more accurate to the control societies, as in the case of evaluation it abandon the individual level and make use of the numbers gathered on each. This culture of comparison is fostered by these self-tracking technologies. Additionally, this constant evaluation could also mean that certain groups, such as those with disabilities, chronic illnesses, or many other things, may face systemic barriers that prevent them from achieving conventional measures of success. By defining success in terms of quantifiable metrics, there is a risk of accentuated stigmatisation of individuals who do not fit within predetermined norms and standards.

Since the globalisation of the economy and the liberalisation of exchanges, the focus on productivity takes a back seat: the main goal of entrepreneurs is now the *competitiveness*. The evaluation becomes a key concept of this competition: there is a shift of its presence, from the work formation or the moment of hiring, to a constant evaluation, maintaining the employee under constant surveillance. This culture of evaluation fosters a sense of competition and comparison, among individuals, as they strive to meet established norms. It creates a pressure to constantly improve themselves, often at the expense of well-being and fulfilment. In this context, the incentive to adopt quantification tools for self-tracking becomes apparent. Individuals internalise this logic of evaluation, seeking to demonstrate their values by the optimisation of their performance. Moreover, the proliferation of digital technologies and Quantified Self applications exacerbates this culture of evaluation by providing individuals with platforms to showcase their achievements. And these applications try to make this process be *fun*. The playful interface of these applications contributes to the phenomenon of “gamification” as explained by professor of gamification Juho Hamari. (HAMARI & al., 2018), that makes the process of self-improvement more enjoyable, interactive, and rewarding. The Quantified Self applications often possess multiple of these gamification features. For example, it is possible to earn achievements and badges through them, or to track the process in general. This phenomenon contributes to attenuate the prescriptive power of these applications.

This reduction of the complexity of an individual through collected numbers and data raises numerous ethical questions. The Quantified Self pushes for self-awareness, but what if this was more of the complete opposite? As said before, self-tracking as it exists today mostly fails to capture the emotions, moods and feelings of an individual, thus making the knowledge of ourselves gained by these tools oversimplify the understanding of human behaviour. By attributing complex descriptions and details solely to quantifiable factors such as steps taken or calories consumed, the Quantified Self movement fails to perceive the influence of socio-cultural, psychological, or even environmental factors on an individual's wellbeing. For example, there exists differences in the ways individuals interact with self-tracking feedback depending on their gender. According to a self-tracking study in 2020, females are more likely than males to perceive feedback as useful, whereas males would more rely on their instincts. So, without clear feedback, females' perceived competences about themselves might suffer more because of their tendency to underestimate themselves. Although males might have a more negative experience with self-tracking, as they have a tendency to overestimate their abilities. (Jin, Halvari, Maehle, Niemiec, 2020) Moreover, the reductionism of the individual with the self-tracking practices may contribute to a loss of self-determination. Indeed, by defining what 'progress' and 'success' is solely in terms of quantifiable metrics and data, individuals internalise external standards of validation, while neglecting their own experiences and values in life. Furthermore, these definition of what 'progress' and 'success' are often related in coaching books, the Quantified Self resembling on some points with the utilitarian philosophy of coaching or the market of "corporate wellness".

2. 3. The neoliberal incentive to ultra-individualisation & the new social links

With the notion of the entrepreneur of the self, neoliberalism forces a sort of ultra-individualisation of society, and undo social links as they existed before. Sociologist Pierre Bourdieu criticise neoliberalism especially for this specific characteristic, as he says neoliberalism fragments collectives to lead into an atomisation of workers, and an erosion of every structure of solidarity in order for the logic of the market to extend into every aspect of social life. (BOURDIEU, 1998: 3) He advances the theory that neoliberalism destroys every collective, so every structure capable of representing an obstacle in the logic of the market, such as with the case for example with work groups by the individualisation of salaries and careers depending on the individual's skills. In a neoliberal society, the individual becomes the finality, and the Quantified Self represents one of its mechanisms to push this narrative. By constantly monitoring and optimising ourselves, the Quantified Self promotes a culture of self-centredness. Making the improvement of the self the priority, the individual tends to then develop a bigger egocentric interest in themselves. The individuals become more preoccupied with themselves, sometimes at the detriment of social interactions. This relentless pursuit of self-improvement leaves little time and little interest into making new relationship and fostering social bonds.

Rather than an atomisation of the social link, the Quantified Self participates to a new kind of culture of creating social links. This new focus on the self through personal development and self-tracking creates, according to sociologist Nicolas Marquis, an "a-sociologism". He defines this notion as a form of relation that links individual and the collective at the same time. Clubs, associations, championships that define a performance framework are concerned by this term, and the self-tracking applications embody yet another form of it. Outside of the individual's own data about their health, it is the performances of others that gives the individual an idea

about the position they have in society, the performative value of every effort put compared to the one of the other users. (MARQUIS, 2014) Obviously, the achievements won in Quantified Self applications are often visible to other people added on the application, such as friends, family or even strangers. From there, it creates an incentive to improve more, thus to play more, for the approval of the individual towards himself, and towards the added relatives. There is an incentive to quantify ourselves more, at least just to feed our ego and see if we walked more than the majority of our friends or not this week. Adding friends and family on these applications allows individuals to judge them more and thus to feel judged ourselves: as said before, this impression fits into the principle of “fitness through shame” of David Pogue, where in this case we are being watched directly and precisely through the Quantified Self applications. On the vast majority of applications using the gamification system, it is possible to compete through leader boards, and to send encouragements to friends whenever their scores are too great or too low. This important element which disrupts the individualising aspect of the Quantified Self practices is called: “interconnectivity”. Users are constantly in link with others, either with family and friends or even with strangers. For example, the Quantified Self application Strava to measure an individual’s activities such as running, cycling and many others, makes possible the interconnexion between every user through posts, followers and likes in the same way as any other social media would. The only exception is that on Strava, rather than other social media applications, the users are not liking posts because they are funny, interesting or from family and friends, but instead because they are impressed that user92837 ran 40 kilometres this week and they wish they would do the same to improve their running capacities.

Conclusion

The Quantified Self is about getting to know ourselves better by measuring data about your bodies and activities. The use of self-tracking devices is intended for self-improvement through self-monitoring of ourselves, to identify and solve specific problems in our daily life. Through monitoring, the long-term goal is to make the Quantified Individual become the Qualified individual through the quantifying tools and practices. Following the creation of the movement in California in 2008, the self-tracking practices have been popularised, in particular due to easier access to new technologies and the datafication of society.

Nevertheless, following our examination through the lenses of Foucault’s concept of disciplinary power and its alignment with neoliberal principles, we have argued that even if the Quantified Self is originally intended for self-knowledge and improvement, it represents a consequence of a bigger system of self-management that reinforces existing power structures. Therefore, it inadvertently contributes to the disciplinary and control powers exerted through internalised control in society and can be considered as one of the mechanisms of the disciplinary societies as conceptualised by Foucault, as well as the control societies theorised by Deleuze. The core argument lies in the convergence of the self-tracking practices used in the Quantified Self with the mechanisms of disciplinary power. In the same way Foucault described the Panopticon as a prison where inmates were constantly observed and finish by internalising this surveillance, the Quantified Self creates its own Panopticon. The individual becomes the architect, the warden, and the prisoner of its prison, by watching and monitoring themselves. In addition, the control societies described by Deleuze also seem to correspond to the mechanism of the Quantified Self, if not more, by its surveillance being in continuum

and the creation of the individual, similar to the digital making of the subject, fragmented and more prone to marketing manipulation. This constant surveillance, driven by the pursuit of optimisation, aligns with the neoliberal ideas which pushes for the individual to become an entrepreneur of the self, which emphasizes self-improvement as the way to success in society. Ultimately, with the normalisation of self-tracking practices considered as “true knowledge”, individuals are expected to continuously optimise themselves, and therefore to contribute to their own commodification. Furthermore, the Quantified Self is a mechanism for neoliberalism to make the individual consider themselves as a finality in itself: the development of the human capital leads to more flexibility, production, and consumerism on the market, and therefore will be enforced into individuals. However, this constant pressure to optimise oneself, aligned with the neoliberal agenda, can lead to a certain social detachment. Due to the popularisation of evaluation in many institutions and the instauration of a sense of competition in education and the workplace, traditional collectives are disappearing to make place for a new kind of social links: interconnectivity.

However, the Quantified Self should not necessarily be seen as a danger nor inherently good or bad, nor an innocent mechanism. This essay’s purpose was to unveil the reason of its apparition, popularisation and mostly institutionalisation throughout society with the lenses of discipline, control, and neoliberalism. Surely the Quantified Self presents many things seen as benefits today besides these points in an individual’s life, further than just monitoring daily life actions. An example could be the lifesavings features some wearable tools possess by measuring the heart rate, thus being able to detect a potential accident and call for help, or its undeniable efficiency for athletes to measure their physical capacities. In the same way, many see its popularisation as the stepping stone of a transhumanism and a danger for humanity, as new technologies are becoming more and more intertwined with human life and privacy invasive, notably with the idea of the exoself mentioned prior. To open a discussion, it would be interesting to see the different discourses and evolution of the technological advancements around the Quantified Self becoming the Qualified Self, and therefore the multiplication of progression towards a potential idea of transhumanism. For example, the American neuro-technology company Neuralink that develops implantable brain-computer interfaces raises serious ethical questions towards the future of human subjectivity and free choice in a close future.

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